

ATPL EASA STUDY GUIDE | 031 - MASS AND BALANCE

Chapter 1: Mass Definitions & Structural Limits

Syllabus Source: CAE Oxford (pp. 1-22) | Pass Target: >= 90% AviationExam

1. THE CORE MASS DEFINITIONS HIERARCHY

In aviation, aircraft mass is divided into precise regulatory categories. Understanding the sequence of additions is vital:

Mass Term	Definition & Constituent Components	Master Equation & Significance
Basic Empty Mass (BEM)	Airframe structure, engines, unusable fuel, full engine oil, hydraulic fluid and permanent standard equipment.	Starting point of aircraft mass; manufacturer weighed.
Dry Operating Mass (DOM)	BEM + Flight Crew & Cabin Crew (+ their baggage) + Catering / Pantry + Removable emergency equipment.	DOM = BEM + Crew + Pantry. Mass of aircraft ready to fly without payload or fuel.
Operating Mass (OM)	DOM + Total Take-Off Fuel.	OM = DOM + Fuel. Does NOT include any passengers or cargo.
Zero Fuel Mass (ZFM)	DOM + Traffic Load (Payload: passengers + baggage + cargo).	ZFM = DOM + Traffic Load. The entire mass of aircraft excluding fuel!
Take-Off Mass (TOM)	ZFM + Total Take-Off Fuel, OR: DOM + Traffic Load + Take-Off Fuel.	TOM = ZFM + Fuel. Mass at brake release on the runway.
Landing Mass (LM)	TOM minus Trip Fuel consumed in flight.	LM = TOM - Trip Fuel.
Ramp / Taxi Mass	TOM + Taxi Fuel (allowance burned during taxi to runway).	Taxi Mass = TOM + Taxi Fuel (max allowable mass at gate).

2. PAYLOAD CONCEPTS: TRAFFIC LOAD VS USEFUL LOAD

- Traffic Load (Payload): The total mass of revenue-earning items: Passengers + Baggage + Cargo + Mail. Formula: Traffic Load = ZFM - DOM.
- Useful Load: The total mass of everything that can be loaded into an empty operating aircraft: Traffic Load + Usable Fuel. Formula: Useful Load = TOM - DOM.

3. STRUCTURAL LIMITS VS PERFORMANCE LIMITS

An aircraft is governed by two distinct sets of mass limits:

- Structural Limits (Maximum Certified): Permanent physical limits established by the manufacturer (MTOM, MZFM, MLM, MTXM). Exceeding these limits risks structural airframe failure (e.g. wing root bending overload at MZFM).
- Performance Limits (Variable / Regulated): Limits calculated for specific runway and environmental conditions (Runway Length Limited TOM, Climb Gradient Limited TOM, Obstacle Limited TOM).
- Golden Rule of Mass Planning: Actual TOM MUST NOT EXCEED the LOWEST of: (1) Structural MTOM, (2) Regulated Performance MTOM, (3) MZFM + Fuel, (4) MLM + Trip Fuel!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Maximum Allowable Take-Off Mass

Exam Problem: MTOM = 70,000 kg, MZFM = 55,000 kg, MLM = 60,000 kg, Take-Off Fuel = 12,000 kg, Trip Fuel = 9,000 kg. What is the maximum regulated TOM? Step 1: Check Structural MTOM: Limit 1 = 70,000 kg. Step 2: Check Wing Bending (MZFM + Fuel): Limit 2 = 55,000 + 12,000 = 67,000 kg. Step 3: Check Destination Landing Mass (MLM + Trip Fuel): Limit 3 = 60,000 + 9,000 = 69,000 kg. Step 4: Golden Rule: Max allowable TOM is the LOWEST of all three limits = 67,000 kg!

4. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Finding Maximum Allowable Take-Off Mass (MTOM)**

SCENARIO (Typical AviationExam Drill): Given the following aircraft data: - Dry Operating Mass (DOM): 35,000 kg - Maximum Structural Take-Off Mass (MTOM): 65,000 kg - Maximum Structural Landing Mass (MLM): 56,000 kg - Maximum Structural Zero Fuel Mass (MZFM): 51,000 kg - Performance Limited TOM for Runway 09: 63,500 kg - Planned Fuel on board at Take-Off: 11,000 kg - Estimated Trip Fuel burn to destination: 4,000 kg QUESTION: What is the MAXIMUM ALLOWABLE TAKE-OFF MASS that respects all limits? SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Check the Direct Take-Off Mass Limits: - Structural MTOM = 65,000 kg - Performance MTOM = 63,500 kg -> (This restricts us to 63,500 kg max directly). Step 2: Check the Zero Fuel Mass Limit constraint on TOM: - Max TOM = MZFM + Take-Off Fuel = 51,000 + 11,000 = 62,000 kg! - (If we took off at 63,500 kg with 11,000 kg of fuel, our ZFM would be 52,500 kg, which EXCEEDS the structural MZFM of 51,000 kg!). Step 3: Check the Landing Mass Limit constraint on TOM: - Max TOM = MLM + Trip Fuel = 56,000 + 4,000 = 60,000 kg! - (If we took off heavier than 60,000 kg, after burning 4,000 kg of trip fuel we would land above the 56,000 kg landing limit!). Step 4: Compare all 3 constrained Take-Off Masses: - Direct Take-off Limit: 63,500 kg - ZFM-Constrained Limit: 62,000 kg - Landing-Constrained Limit: 60,000 kg FINAL ANSWER: The Maximum Allowable Take-Off Mass is 60,000 kg! (Limited by Landing Mass).

[!] EXAM TRAP (AVIATIONEXAM) - The Wing-Root Bending / MZFM Exam Trap

- Why does MZFM exist? All payload is carried inside the fuselage, while fuel is carried inside the wings. In flight, wing fuel provides an upward relieving moment. When fuel burns down to zero, the entire heavy fuselage load is supported exclusively by the wing roots! Exceeding MZFM breaks the wing roots, even if you are well below MTOM!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the official definition of DRY OPERATING MASS (DOM)?**

- [A]: The mass of the airplane empty of all fuel and oil
- [B]: Basic Empty Mass (BEM) PLUS crew, crew baggage, catering/pantry, and removable emergency equipment
- [C]: Zero Fuel Mass minus payload
- [D]: Maximum Take-Off Mass minus fuel

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. DOM is the total mass of the aircraft ready for a specific type of operation, including all crew, their personal baggage, and catering, but excluding all usable fuel and traffic load (payload).

Q2: If an aircraft's Maximum Zero Fuel Mass (MZFM) is 45,000 kg, Dry Operating Mass is 28,000 kg, and planned

- [A]: 25,000 kg
- [B]: 17,000 kg
- [C]: 9,000 kg
- [D]: 12,000 kg

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Traffic Load = ZFM - DOM. The maximum possible ZFM is MZFM (45,000 kg).
Therefore: Max Traffic Load = 45,000 - 28,000 = 17,000 kg.

Q3: Which component is included in the BASIC EMPTY MASS (BEM) of an aircraft?

- [A]: Flight crew baggage
- [B]: Usable fuel in the wing tanks
- [C]: Unusable fuel, engine oil, hydraulic fluid, and standard permanent equipment
- [D]: Passenger catering supplies

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Basic Empty Mass includes the complete structural airframe, engines, standard fixed equipment, plus unusable fuel and full operating fluids (engine oil and hydraulic systems). Crew and catering are added to BEM to form DOM.